

Module 08: Planning in Robotics — Bio-Inspired Planning

Learning Objectives

1. Explain how **biological planning mechanisms** (mental simulation, hierarchical goal decomposition, anticipatory behavior) inspire robotic planning architectures.
2. Analyze the role of **Expected Free Energy minimization** as a unified framework for bio-inspired robotic planning.
3. Apply bio-inspired planning strategies to achieve **flexible, robust goal pursuit** in unstructured environments.

Introduction

Biological organisms plan at multiple timescales: a squirrel plans a cache route across the park over minutes, a crow plans a multi-step tool use sequence over seconds, and a cat plans a pounce trajectory over milliseconds. These planning capacities inspire robotic planning systems that combine the flexibility of hierarchical goal decomposition with the robustness of reactive behavior.

Key Concepts

1. Expected Free Energy Planning

In Active Inference, planning is policy selection by evaluating **Expected Free Energy** (EFE) for each candidate policy over a finite planning horizon:

$$G(\pi) = \Sigma [\text{-Pragmatic Value}(\pi, t) - \text{Epistemic Value}(\pi, t)]$$

- **Pragmatic value:** How close does policy π bring the agent to its preferred observations at time t ?
- **Epistemic value:** How much does policy π reduce the agent's uncertainty about the world at time t ?

Bio-inspired robots plan by evaluating EFE across a policy tree — exploring promising branches and pruning unpromising ones. The epistemic-pragmatic decomposition naturally handles the exploration-exploitation trade-off that plagues standard planning algorithms.

2. Hierarchical Goal Decomposition

Biological planning is hierarchical — high-level goals are decomposed into sub-goals, sub-goals into actions:

- "Build a nest" → "Gather materials" + "Weave structure" + "Line interior"
- "Gather materials" → "Fly to source" + "Select appropriate branch" + "Break off segment" + "Carry back"

Bio-inspired robotic planners replicate this hierarchy through **hierarchical MDP/POMDP models** where higher levels set goals for lower levels, and lower levels report completion back up. Each level performs Active Inference at its own timescale.

3. Anticipatory Behavior

Bio-inspired robots use **anticipatory control** — acting based on predicted future states rather than current observations:

- A robotic catcher predicts the ball's trajectory and moves its hand to the future intercept point — not the current ball position
- A warehouse robot predicts human workers' paths and proactively avoids future collision zones
- An agricultural robot predicts crop growth stages and schedules interventions accordingly

Anticipatory behavior is planning compressed to the motor level — the generative model's future predictions directly drive current actions.

4. Planning Under Uncertainty: Information-Gathering Strategies

Biological organisms actively gather information before committing to a plan:

- A crow inspects a novel puzzle from multiple angles before attempting a solution (visual information gathering)
- A rat pauses at a T-maze junction as hippocampal place cells "preview" both arms (vicarious trial and error)
- A cat stalks prey using a circuitous approach that provides visual access from multiple angles before committing to a pounce

Bio-inspired robots replicate these strategies through **epistemic actions** — deliberately information-seeking movements that reduce planning uncertainty before executing the main plan.

Applications

- **Agricultural harvesting robot:** A bio-inspired harvesting robot uses hierarchical planning (field coverage → row traversal → individual plant harvesting), anticipatory behavior (predicting fruit ripeness from visual features to optimize harvest timing), and epistemic inspection (pausing to image fruit from multiple angles before selecting the grasp strategy).
- **Collaborative construction robot:** A robot building structures alongside humans uses anticipatory planning (predicting the human's next action and pre-positioning materials) and hierarchical decomposition (mission → wall segment → individual brick placement) — combining bio-inspired flexibility with precise execution.

Conclusion

Bio-inspired planning — Expected Free Energy minimization, hierarchical decomposition, anticipatory behavior, and information-gathering strategies — gives robots the capacity for flexible, robust goal pursuit in unstructured environments. This completes the Bio-Inspired Design unit.